

ASSIGNMENT 2

Covariance, VaR, and Copulas

COURSE	FinTech 545 - Quantitative Risk Management
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Approach

Each problem is organized as Predict, Fit, and Reconcile. Predictions are based only on the requested exploratory evidence; fitted results then test those predictions. Risk numbers use positive dollar losses and the course finite-sample VaR convention.

Headline results: mismatched pairwise histories create a negative portfolio variance; a volatility regime shift dominates EW VaR; diversification raises 5% VaR but lowers ES for defaultable bonds; a Student-t copula decisively captures joint tails; and positive residual correlation raises long-long risk while reducing long-short risk.

1. Correlations from Mismatched Histories

Predict. Observation overlap and PSD risk

Pairwise estimation can produce a non-PSD matrix because every cell may use a different sample. Complete-case estimation standardizes one common data matrix, so its correlation matrix is a Gram matrix and is PSD apart from roundoff. Because IDX is almost a linear combination of A, B, C, and D, I predicted a near-zero smallest eigenvalue and unusual sensitivity to even small pairwise inconsistencies.

	A	B	C	D	IDX
A	242	242	221	93	242
B	242	242	221	93	242
C	221	221	228	83	228
D	93	93	83	96	96
IDX	242	242	228	96	250

Only **81** days contain all five series. Before calculating correlations, I expected pairs containing D to be least reliable: C-D has 83 joint days, A-D and B-D have 93, and D-IDX has 96.

Fit. Complete-case and pairwise matrices

Complete-case correlation (81 rows)

	A	B	C	D	IDX
A	1.000	0.397	0.247	0.414	0.871
B	0.397	1.000	0.317	0.236	0.671
C	0.247	0.317	1.000	0.185	0.565
D	0.414	0.236	0.185	1.000	0.572
IDX	0.871	0.671	0.565	0.572	1.000

Complete-case eigenvalues: 0.001506, 0.548345, 0.693320, 0.870441, 2.886388. Ordinary Cholesky succeeds. The minimum eigenvalue is positive but small, confirming the near-singularity prediction.

Pairwise correlation

	A	B	C	D	IDX
A	1.000	0.469	0.431	0.438	0.870
B	0.469	1.000	0.463	0.277	0.733
C	0.431	0.463	1.000	0.185	0.714
D	0.438	0.277	0.185	1.000	0.606
IDX	0.870	0.733	0.714	0.606	1.000

Pairwise eigenvalues: -0.011072, 0.471422, 0.528367, 0.857381, 3.153902. The minimum is **-0.01107185**, so ordinary Cholesky fails. Using each series' full-history sample standard deviation and $\Sigma = \text{diag}(\text{sd}) R \text{diag}(\text{sd})$, the tracking weights (-0.4,-0.3,-0.2,-0.1,1) produce variance **-0.0000017994**.

Full-history sample SDs used in the reconstruction are A=0.016502, B=0.011919, C=0.018324, D=0.022829, IDX=0.012281.

PSD means $w' \Sigma w \geq 0$ for every weight vector w . The specified tracking vector gives $w' \Sigma w < 0$, so Σ is not PSD; equivalently, its negative eigenvalue guarantees a direction with a negative quadratic form. This is a matrix inconsistency, not an economic variance.

Fit. Correlation repairs

Method	Min eigenvalue	Frobenius distance	Max cell move	Tracking variance
Rebonato-Jackel	1.47e-16	0.01618210	0.00754786	6.80729771e-07
Higham	0 (roundoff -6.24e-15)	0.01552000	0.00704463	6.78764948e-07

Rebonato-Jackel clips negative eigenvalues and rescales the spectral root to a unit diagonal. Higham uses alternating PSD and unit-diagonal projections with Dykstra correction; it converged in 40 iterations. Both repairs restore a small positive tracking variance and are very close to one another. Higham's reported -6.24e-15 minimum eigenvalue is floating-point zero, not a material PSD failure.

Reconcile. What moved, and what mattered most

Entry	Pairwise	Higham	Signed move
A-IDX	0.869584	0.862540	-0.007045
C-IDX	0.713575	0.708810	-0.004765
B-IDX	0.732867	0.728687	-0.004180
D-IDX	0.605691	0.601964	-0.003727
A-C	0.430888	0.433165	0.002277

The largest Higham move is A-IDX, even though that pair has 242 observations; C-D has only 83 but moves less. Higham was not given observation counts. It uses global eigenstructure and equal Frobenius cost per cell, changing the collection of entries that most cheaply removes the negative direction.

The complete-case/pairwise distance is 0.423935, or 27.32 times the Higham repair distance. For these data, estimator choice matters much more than choosing between the two repairs.

2. A Volatility Estimate After the Regime Changed

Predict. The recent regime should dominate EW risk

I computed arithmetic returns, removed the full-sample mean, and inspected the plot before calculating VaR. Volatility rises sharply near day 461, so I classified the last 40 returns as the recent regime. I predicted, from smallest to largest: equal-weight Normal, historical, Student-t, EW(0.97), EW(0.94). Full-sample skew and heavy tails should put historical and Student-t beyond a Normal benchmark, with the exact order of those two least certain. The EW measures should be largest because recent volatility is higher, and $\lambda=0.94$ reacts most quickly.

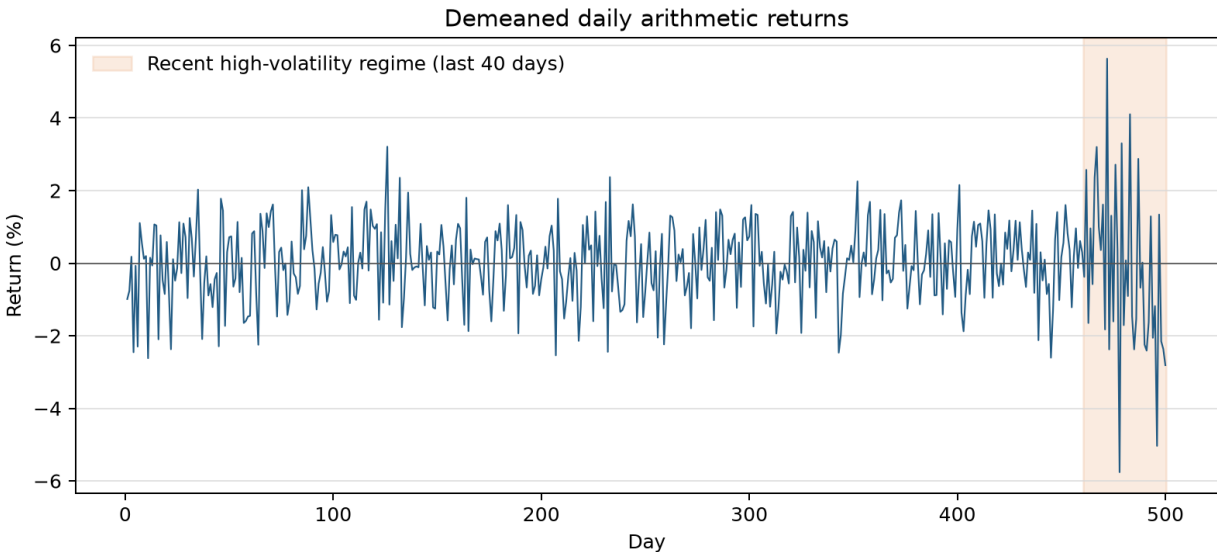


Figure 1. Demeaned arithmetic returns. The shaded final 40 days define the visually selected high-volatility regime.

Mean	Variance	SD	Skewness	Excess kurtosis
-2.22e-19	0.00014190	0.011912	-0.156	2.166

Excess kurtosis of 2.166 is incompatible with a single stationary Normal regime as a credible description. The prediction is about stationarity, not merely whether a Normal sample could ever show a nonzero sample moment.

Predict (continued). Effective memory of the two decay factors

lambda	Effective n	Half-life	Weight on last 40	EW SD	VaR
0.94	32.33	11.20 days	91.6%	0.02353	\$38,707.38
0.97	65.67	22.76 days	70.4%	0.02105	\$34,619.08

The plot suggests 40 recent days are different. Lambda=0.94 assigns 91.6% of its total weight to them, compared with 70.4% for lambda=0.97, reinforcing the prediction that 0.94 will produce the larger risk estimate.

Fit. Fitted t and five VaRs

Rank	Method	5% one-day VaR (loss)
1	Student-t MLE	\$18,872.44
2	Equal-weight Normal	\$19,593.66
3	Historical	\$20,574.46
4	EW Normal 0.97	\$34,619.08
5	EW Normal 0.94	\$38,707.38

The Student-t MLE is location=0.00018164, scale=0.01028999, and nu=8.2699. Scale is not SD; the implied SD is 0.01181775. Historical VaR uses the course RiskStats order-statistic convention, here the 25th smallest return.

Reconcile. A regime mixture, not one persistent heavy tail

Regime	Days	SD	Skewness	Excess kurtosis
Earlier	460	0.010178	-0.220	-0.202
Recent	40	0.024289	0.210	0.030

Recent SD is 2.386 times earlier SD, while each regime's excess kurtosis is near zero. Mixing their variances implies excess kurtosis about 2.572. The fitted t is mainly a one-distribution approximation to a volatility-regime mixture.

The EW VaR gap is \$4,088.30; the VaR-equivalent SEs are \$4,813.42 and \$3,020.84. Thus the gap is 0.85 and 1.35 times those individual SEs, and only 0.72 times the conservative root-sum-square noise scale. It is not convincingly larger than estimation noise. **Opinion:** I would use 0.94 for a responsive trading limit and 0.97 for a more stable capital number.

Reconciliation: EW(0.94) > EW(0.97) agrees with the prediction, but the actual full-sample order is Student-t < Normal < historical, not Normal < historical < Student-t. A fitted t is both more peaked in the center and heavier in the far tails. At 5% its smaller fitted scale dominates; much farther out, its polynomial tail would dominate.

3. When Diversification Raises VaR

Predict. A probability jump crosses the 5% cutoff

The return distributions have a large nondefault profit cluster and a separate default loss cluster. Both bonds are extremely left-skewed and leptokurtic, so Normal VaR will smooth a discrete default jump into an ordinary standard deviation and cannot describe the actual tail mechanism.

Bond	Mean	Variance	Skewness	Excess kurtosis
A	0.083709	0.018493	-4.823	22.094
B	0.085011	0.017451	-4.916	23.087

A loses more than 20% in 406 scenarios (4.06%); B in 393 (3.93%); at least one does in 782 (7.82%); both do in only 17 (0.17%). Each standalone default rate is below 5%, but the union rate exceeds 5%, so diversification should raise 5% VaR. Only 17 cases are joint defaults: the diversified worst 5% is therefore dominated by one \$1m bond defaulting, while the concentrated tail exposes the full \$2m to A's default. I predicted lower diversified ES.

Problem 3: diversification moves default losses across the 5% threshold

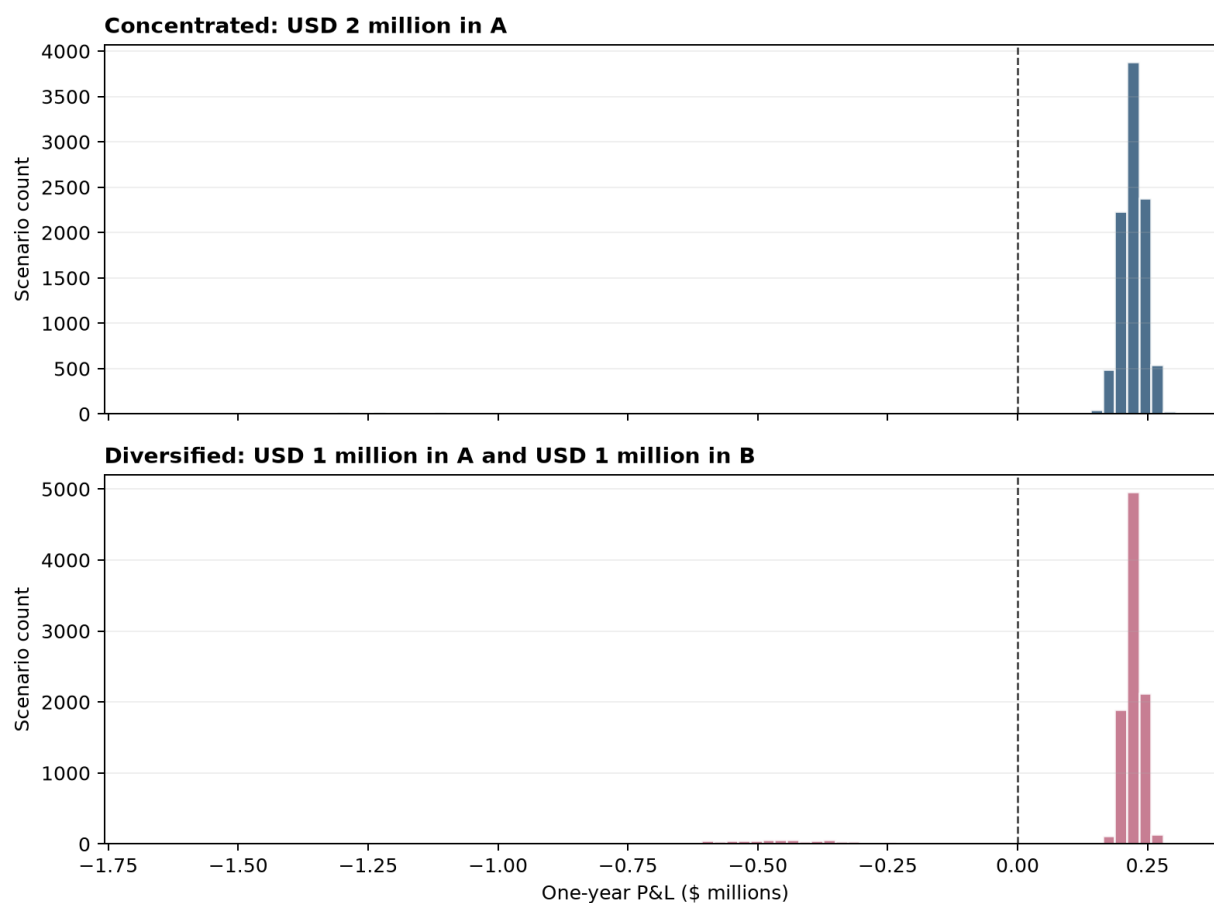


Figure 2. Concentration has fewer but more severe default states; diversification creates a more frequent one-default loss region.

Fit. Historical and Normal risk

Position	Hist VaR 5%	Hist ES 5%	Normal VaR	Hist VaR 1%	Hist ES 1%
\$1m A	-\$84,530.68	\$443,628.30	\$139,970.62	\$650,545.13	\$709,547.78
\$1m B	-\$86,011.21	\$419,587.29	\$132,276.77	\$641,248.09	\$700,805.34
\$2m A	-\$169,061.35	\$887,256.60	\$279,941.24	\$1,301,090.25	\$1,419,095.55
\$1m A + \$1m B	\$414,766.85	\$541,997.24	\$143,466.25	\$598,237.15	\$716,407.69

Negative 5% VaR for a standalone bond means its 5th-percentile scenario is still a profit; it is not clipped to zero. ES remains large because it averages the rarer default states beyond that quantile.

Reconcile. VaR fails subadditivity; ES does not

At 5%, combined VaR is \$414,766.85, while the two standalone VaRs sum to -\$170,541.89. VaR subadditivity fails by \$585,308.74. Combined ES is \$541,997.24, below the standalone sum \$863,215.59; ES has \$321,218.35 of subadditivity slack.

Both portfolio-level predictions are confirmed: diversification raises VaR from -\$169,061.35 to \$414,766.85, but lowers ES from \$887,256.60 to \$541,997.24.

Normal VaR favors diversification (\$143,466.25 versus \$279,941.24), but for the wrong reason: it rewards lower variance while missing the probability jump. At 1%, historical VaR reverses: diversified \$598,237.15 versus concentrated \$1,301,090.25. The 1% cutoff lies inside each bond's default region, so both portfolios now reveal default risk and splitting notional reduces severity. VaR's failure is confidence-level dependent.

4. Gaussian or Student-t Copula

Predict. Margins and empirical tail clustering

Series	Mean	Variance	Skewness	Excess kurtosis	Prediction
X1	0.000257	0.0001415	-0.034	-0.117	Normal
X2	-0.000252	0.0002614	-0.615	7.125	Student-t
X3	0.000890	0.0001004	0.074	2.874	Student-t

X2's kurtosis of 7.125 looks exceptional. Observation 952 contains X2=-14.291% and is also extreme for the other margins. Removing it lowers X2 kurtosis to 1.725 and X3 kurtosis to 1.731; both remain positive, so the Student-t predictions do not depend on one point alone.

Pair	Observed lower	Observed upper	Independence
X1-X2	6	8	0.625
X1-X3	7	12	0.625
X2-X3	6	9	0.625

Independence expects only 0.625 days per pair and side; observed counts range from 6 to 12. The rank plots show a broadly related but diffuse middle, while the corner clusters look unusually dense. Together with the common outlier, that suggested dependence beyond ordinary middle correlation, so I predicted that the Student-t copula would win.

Problem 4: marginal rank pairs (independence expects 0.625 days in each joint tail)

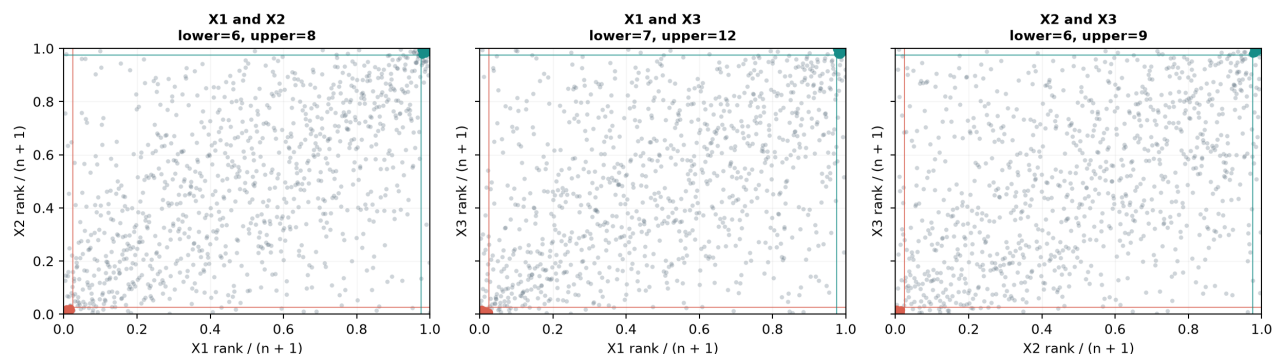


Figure 3. Pairwise pseudo-observations rank/(n+1); highlighted corners show joint 2.5% tail days.

Fit. Marginal model selection

Series	Normal AICc	t nu	t AICc	Selected	AICc advantage
X1	-6022.54	Normal limit	-6020.52	Normal	2.01
X2	-5408.49	5.272	-5518.18	Student-t	109.69
X3	-6365.69	6.095	-6428.02	Student-t	62.32

X1's fitted t has effectively infinite degrees of freedom, so it is the Normal limit and cannot justify an extra parameter. AICc selects Normal for X1 and Student-t for X2 and X3.

Series	Selected fit	Location	Scale	nu
X1	Normal	0.0002566	0.0118882	n/a
X2	Student-t	-0.0001760	0.0125440	5.272
X3	Student-t	0.0007456	0.0081808	6.095

For copula fitting, I transformed each raw return with its selected fitted marginal CDF. These parametric uniforms differ from the rank/(n+1) values used only for the exploratory plot and counts.

Pair	Kendall tau	$\rho = \sin(\pi \cdot \tau / 2)$
X1-X2	0.39645	0.58326
X1-X3	0.33543	0.50285
X2-X3	0.30745	0.46438

Copula	Log likelihood	Parameters	AICc	BIC
Gaussian	350.440	3	-694.856	-680.157
Student-t	451.999	4	-895.959	-876.368

The Kendall-tau correlation matrix is already positive definite. The Student-t copula profiles to nu=3.6599. It wins AICc by 201.102 and BIC by 196.211, decisive evidence for tail dependence.

Model	VaR 5%	ES 5%	VaR 1%	ES 1%
Historical	\$47,939.19	\$65,698.80	\$70,092.00	\$102,424.22
Gaussian copula	\$49,796.41	\$67,320.32	\$77,451.22	\$96,047.44
Student-t copula	\$49,128.30	\$68,953.97	\$79,780.81	\$102,870.31

Reconcile. What the t copula fixes

Pair	Obs L	G L	t L	Obs U	G U	t U
X1-X2	6	6.28	9.57	8	5.98	9.72
X1-X3	7	5.01	8.35	12	4.65	8.40
X2-X3	6	4.88	8.22	9	4.03	7.36

Across the six pair-tail cells, the data contain 48 joint days, versus 30.83 implied by the Gaussian and 51.62 by the Student-t. Thus the t copula is much closer to the observed aggregate. Its 5% VaR falls by only \$668.12; the 5% cutoff is not deep enough for simultaneous extremes to dominate. The largest movement is 1% ES, which rises by \$6,822.86, because ES averages losses beyond an already extreme cutoff.

Daily log-density differences sum to 101.5591. The 784 days on which every empirical rank lies between 5% and 95% contribute 49.8003, or 49.04% of the total. Information criteria score every observation, not only hand-picked tail days.

For the most correlated pair, X1-X2 ($\rho=0.583$), the fitted Student-t lower and upper tail-dependence coefficient is 0.3219. The Gaussian value is exactly zero for correlation below one.

5. Model-Based Simulation and Residual Correlation

Predict. The covariance cross-term changes sign

If A and B share an omitted industry exposure, $\text{Cov}(e_A, e_B)$ is positive. In $\text{Var}(w_A r_A + w_B r_B)$, the cross-term is $2 w_A w_B \text{Cov}(r_A, r_B)$. P1 has same-sign weights, so setting residual covariance to zero must understate its VaR. P2 has opposite-sign weights, so the same shortcut removes a hedge and must overstate VaR.

Fit. OLS and two residual simulations

The 501 price observations produce 500 arithmetic returns for each series.

Stock	alpha	beta	Residual SD
A	0.00020501	1.035278	0.01288830
B	0.00031477	0.935417	0.01153844

Residual correlation is **0.698307**, large and positive. The assignment assumes zero expected returns, so I report the fitted intercepts as OLS diagnostics but omit alpha from the risk simulation; market and residual draws are centered. Full and diagonal-residual simulations use the same 100,000 Gaussian draws.

Portfolio	Full residual VaR	Independent VaR	Independent - full	Delta-normal
P1: long A + long B	\$50,220.00	\$44,418.12	-\$5,801.88	\$50,414.62
P2: long A - short B	\$15,793.73	\$28,491.54	\$12,697.81	\$15,833.60

Problem 5: positive residual correlation raises P1 risk and lowers P2 risk

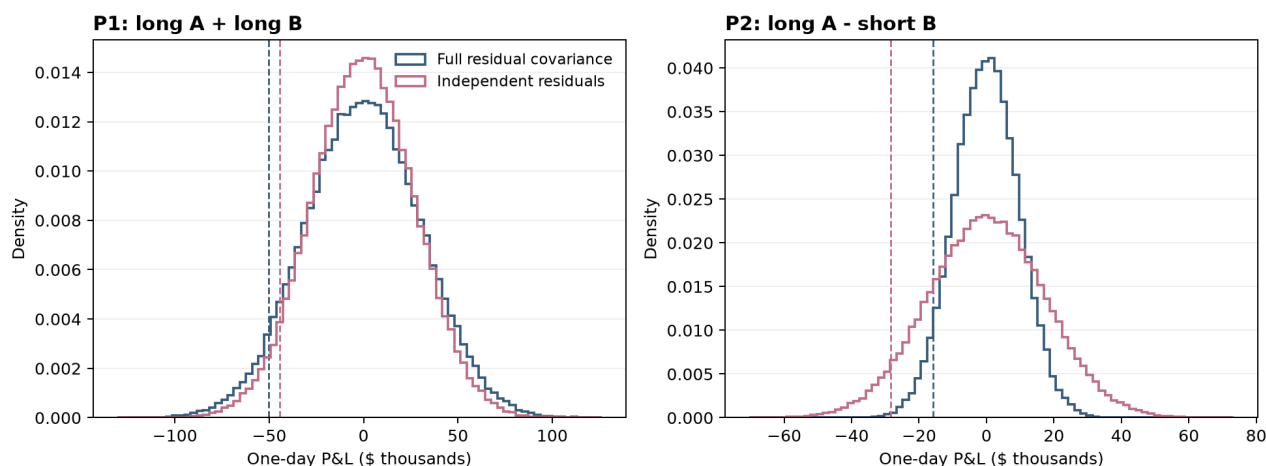


Figure 4. Ignoring positive residual correlation narrows P1 but widens the long-short P2 distribution.

Reconcile. The shortcut removes risk from P1 and a hedge from P2

The direction matches the prediction. Independence lowers P1 VaR by \$5,801.88 (11.55%), but raises P2 VaR by \$12,697.81 (80.40%). P2 is more exposed because common residual movements hedge a long-short spread; setting covariance to zero removes that hedge.

The factor reconstruction and direct stock covariance agree to $1.63\text{e-}19$. Under OLS, residuals are sample-orthogonal to the market, so $\text{Cov}(r) = \text{beta beta}' \text{Var}(\text{MKT}) + \text{Sigma}_e$. With joint Normal components and linear P&L, factor simulation and delta-normal describe the same distribution. The full simulation is \$194.62 below delta-normal for P1 and \$39.87 below it for P2; these small differences are Monte Carlo quantile noise. They would separate with nonlinear positions, heavy tails, heteroskedasticity, time-varying betas, or residual dependence on the market.

Methods and Reproducibility

All calculations are generated from the five supplied CSV files by `run_assignment.py`. The reusable `risk545` package contains the model logic; `build_report.py` reads the generated JSON and figures, so the written numbers and code output share one source.

Quantity	Convention
Returns	Arithmetic $P[t]/P[t-1]-1$
Moments	Sample variance (n-1); corrected skewness and excess kurtosis
Historical VaR	Course RiskStats floor/ceil one-based order-statistic average
Historical ES	Negative mean P&L; at or below the VaR cutoff, including ties
AICc/BIC	Every estimated margin or copula parameter counted
Exploratory ranks	Average rank/(n+1), strictly inside (0,1), for plots and tail counts
Copula uniforms	Selected fitted marginal CDF applied to each raw return
Simulation	100,000 draws; master seed 545

Reproduction: create a Python environment, install `requirements.txt`, run `python run_assignment.py --seed 545`, then run `python -m pytest -q`. The README contains the complete commands.